Relational Algebra

Chapter 4

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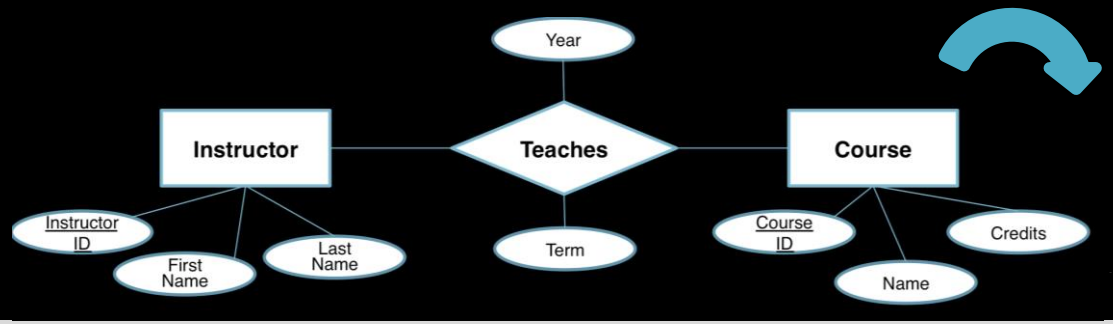
Last Time

ER to Relational Model

- Conversion of
  - Simple constructs: Entities, relationships, attributes
  - Complex constructs: ISA, weak entities
  - Integrity constraints: keys
- Triggers and constraints for consistency
  - Constraints are easier to understand
  - Triggers are more powerful



Instructor ID	First Name	Last Name
394953	John	Smith
454544	Sara	King
439849	Alex	Dee
....	....	....
....	....	....

Instructor ID	Course ID	Year	Term
454544	E302	2009	F
394953	C210	2010	W
439849	M184	2010	F
....	....	....	....
....	....	....	....

Course ID	Name	Credits
M184	Calculus	3
C210	Physics	4
E302	Algorithms	4
....	....	....
....	....	....

Tea Time: Keemun Tea

- 祁門紅茶
- Black tea originally from Anhui province
- slow withering and oxidation processes
- gentle, malty, floral aromas
- Tastes almost like unsweetened cocoa

Random Note About UMich Alumni

3rd largest alumni network in the world (~700K):

- 1 U.S. president,
- 2 dozen dozen governors,
- 3 Supreme Court justices,
- 10 Nobel award winners

But **more importantly** a lot of DB stars!

- Wesley Chu (MS'61),
- E.F. Codd (PhD'65), father of modern DBs
- Mike Stonebraker (PhD'71): **Ingres**, **Postgres**, ..., the most prolific DB systems builder in history
- S. Bing Yao (PhD'74),
- David DeWitt (PhD'76),
- Sham Navathe (PhD'76),
- Francois Bancilhon (PhD'76),
- Kevin O'Connor (BSE'83),
- and Chin-Wan Chung (PhD'83).

Formal Query Languages

- Foundation for commercial query languages like SQL
- Two types
  - Declarative: Relational Calculus
    - Describe what a user wants, rather than how to compute it
  - Imperative: Relational Algebra
    - Operational, very useful for representing execution plans
- Query Languages **!=** programming languages
  - QLs not expected to be "Turing complete"
  - QLs not intended to be used for complex calculations
  - QLs support easy, efficient access to large data sets

Understanding Algebra ~~& Calculus~~ is key to understanding SQL query processing!

One more random thing: Greek Alphabet

a.k.a. “Let’s practice our Greek alphabet!”

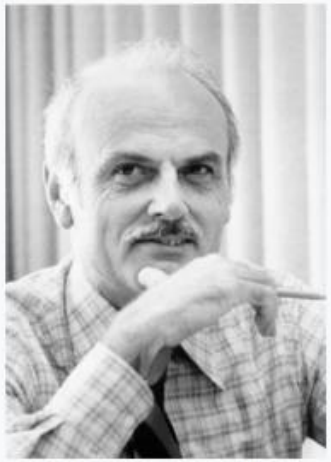
Greek Alphabet

Αα	Ββ	Γγ	Δδ	Εε	Ζζ
Alpha	Beta	Gamma	Delta	Epsilon	Zeta
Ηη	Θθ	Ιι	Κκ	Λλ	Μμ
Eta	Theta	Iota	Kappa	Lambda	Mu
Νν	Ξξ	Οο	Ππ	Ρρ	Σσ
Nu	Xi	Omicron	Pi	Rho	Sigma
Ττ	Υυ	Φφ	Χχ	Ψψ	Ωω
Tau	Upsilon	Phi	Chi	Psi	Omega

Relational Algebra Preliminaries

- Query:
  - Input: one or more relations
  - Output: one relation
    - May produce different results for different input instances
    - But schema of the result is fixed

Edgar "Ted" Codd



Born	Edgar Frank Codd 19 August 1923[1][2] Fortuneswell, Dorset, England
Died	18 April 2003 (aged 79) Williams Island, Aventura, Florida, USA
Alma mater	Exeter College, Oxford University of Michigan

Blame your own alma mater

Relational Operators

- Selection σ: Selects a subset of rows from relation
- Projection π: Deletes unwanted columns from relation

1 reln

- Cross-product $\times$: Allows us to combine two relations
- Set-difference $-$: Tuples in reln. 1, but not in reln. 2
- Union $\cup$: Tuples in reln. 1 or in reln. 2

2 reln

- Additional operations (constructed from basic ops – not essential, but very useful):
  - Intersection, **Join**, Division, Renaming
- Because algebra is closed, we can **compose** operators

Selection

- Retrieve **rows** that satisfy a logical condition

$$\sigma_{condition}(relation)$$

- Conditions are boolean combinations (∧, ∨) of terms
- Each term has of the following forms:
  - Attr1 <oper> Attr2
  - Attr <oper> value

  where <oper> in {>, ≥, =, ≤, <, <>}

Athlete

aid	name	sport	country
1	Mary Lou Retton	gymnastics	USA
2	Jackie Joyner-Kersee	track	USA
3	Michael Phelps	swimming	USA
4	Johann Koss	skating	Norway
5	Natalie Coughlin	swimming	USA
6	Gabby Douglas	gymnastics	USA

$$\sigma_{sport='gymnastics' \wedge country='USA'}(Athlete)$$

Q: what is the degree & cardinality of the above relation?

A) 4, 2
B) 2, 2
C) 2, 5
D) 4, 5

Selection

- Retrieve **rows** that satisfy a logical condition

$$\sigma_{condition}(relation)$$

- Conditions are boolean combinations (∧, ∨) of terms
- Each term has one of the following forms:
  - Attr1 <oper> Attr2
  - Attr <oper> value

  } where <oper> in {>, ≥, = ≤, <}

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3	Michael Phelps	swimming	USA
4	Johann Koss	skating	Norway
5	Natalie Coughlin	swimming	USA
6	Gabby Douglas	gymnastics	USA

$$\sigma_{sport='gymnastics' \wedge country='USA'}(Athlete)$$

aid	name	sport	country
1	Mary Lou Retton	gymnastics	USA
6	Gabby Douglas	gymnastics	USA

Correct Answer: A
Degree 4, cardinality 2

Projection

- Chooses some columns of interest
- Delete attributes that are not in projection list
- Remove **duplicates**!
  - Remember classical relational model is a **SET**

Athlete

aid	name	sport	country
1	Mary Lou Retton	gymnastics	USA
2	Jackie Joyner-Kersee	track	USA
3	Michael Phelps	swimming	USA
4	Johann Koss	skating	Norway
5	Natalie Coughlin	swimming	USA
6	Gabby Douglas	gymnastics	USA

$$\pi_{sport,country}(Athlete)$$

sport	country
gymnastics	USA
track	USA
swimming	USA
skating	Norway

Set Operations: Union (∪), Intersection(∩), Set-Difference (-)

- **Input:** Two union-compatible relations
  1. Same number of attributes
  2. Attributes have the same domain from left to right
  - Corresponding attrs do **NOT** have to have the same name
- **Field names of result:** uses name from the FIRST input

name
Mary Lou Retton
Jackie Joyner-Kersee
Michael Phelps
Johann Koss
Natalie Coughlin
Gabby Douglas

−

AthleteName
Mary Lou Retton
Jackie Joyner-Kersee
Apollo Ono
Natalie Coughlin
Bode Miller

=

name
Michael Phelps
Johann Koss
Gabby Douglas

Cross-Product (Cartesian Product)

- **Result Schema:**
  - All fields from both relations, names inherited if possible

Athlete

aid	name	sport	country
1	Mary Lou Retton	gymnastics	USA
2	Jackie Joyner-Kersee	track	USA

×

Venue

vid	venue
1	Los Angeles
2	Barcelona

=

aid	name	sport	country	vid	venue
1	Mary Lou Retton	gymnastics	USA	1	Los Angeles
2	Jackie Joyner-Kersee	track	USA	1	Los Angeles
1	Mary Lou Retton	gymnastics	USA	2	Barcelona
2	Jacki Joyner-Kersee	track	USA	2	Barcelona

Cross-Product (Cartesian Product)

- **Result Schema:**
  - All fields from both relations, names inherited if possible
- Attrs that have the same name will be unnamed (& referred to by their position) unless we use renaming operator

Athlete

aid	name	sport	venue
1	Mary Lou Retton	gymnastics	USA
2	Jackie Joyner-Kersee	track	USA

×

Venue

vid	venue
1	Los Angeles
2	Barcelona

=

aid	name	sport	(venue)	vid	(venue)
1	Mary Lou Retton	gymnastics	USA	1	Los Angeles
2	Jackie Joyner-Kersee	track	USA	1	Los Angeles
1	Mary Lou Retton	gymnastics	USA	2	Barcelona
2	Jacki Joyner-Kersee	track	USA	2	Barcelona

Rename Operator

Syntax: ρ (newRelationName, expr)

ρ (newRelName(oldAttr1→newAttr, …), expr)

ρ (oldAttr1→newAttr1, …, expr)

- Use cases:
  - Shorthand ρ (R, $\sigma_{sport='track'}$(Athlete))
  - Self-joins ρ (R1, Athlete) x ρ (R2, Athlete)
  - Same attr names: ρ (aid→aid1, Athlete) x ρ (aid→aid2, Compete)

Example: ρ (1→id, name→fullname, $\pi_{aid,name,\ sport}$(Athlete))

aid	name	sport	country
1	Mary Lou Retton	gymnastics	USA
2	Jackie Kersee	track	USA

id	fullname	sport
1	Mary Lou Retton	gymnastics
2	Jackie Kersee	track

Derived Operators: Joins

- Most common way of combining information from two tables
- **Conditional join or Θ-join**
  - $R \bowtie_c S = \sigma_c(R \times S)$, where *c* is a condition
- **Equijoin**
  - Join condition consists only of equalities
- **Natural Join** $R \bowtie S$
  - Special case of equijoin but with implicit equality condition for identical column names AND drops the duplicate column from second relation
- Despite equivalence, joins faster than cross-product!

Example

Writing Queries in RA: Example 1

```
Sailors (sid, sname, rating, age)
Reserves (sid, bid, day)
Boats (bid, bname, color)
```

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Reserves

sid	bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Find names of sailors who have reserved boat #103

Example

Find names of sailors who have reserved boat #103

Solution 1: $\pi_{sname}((\sigma_{bid=103} Reserves) \bowtie Sailors)$

Solution 2:

$$\rho\ (Temp1,\ \sigma_{bid=103} Reserves)$$

$$\rho\ (Temp2,\ Temp1 \bowtie Sailors)$$

$$\pi_{sname}(Temp2)$$

Solution 3: $\pi_{sname}(\sigma_{bid=103}(Reserves \bowtie Sailors))$

Example

Writing Queries in RA: Example 2

Find names of sailors who have reserved a red boat

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Reserves

sid	bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Example

Find names of sailors who have reserved a red boat

$$\pi_{sname}((\sigma_{color='red'} Boats) \bowtie Reserves \bowtie Sailors)$$

An equivalent solution:

$$\pi_{sname} (\pi_{sid} ((\pi_{bid} \sigma_{color='red'} Boats) \bowtie Res) \bowtie Sailors)$$

The query optimizer chooses from the (equivalent) expressions and chooses the most efficient one.

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Reserves

sid	bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Example

Writing Queries in RA: Example 3

Explain what this query does in plain English?

Think about this later

$$\pi_{sname}(Reserves \bowtie Sailors)$$

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Reserves

sid	bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

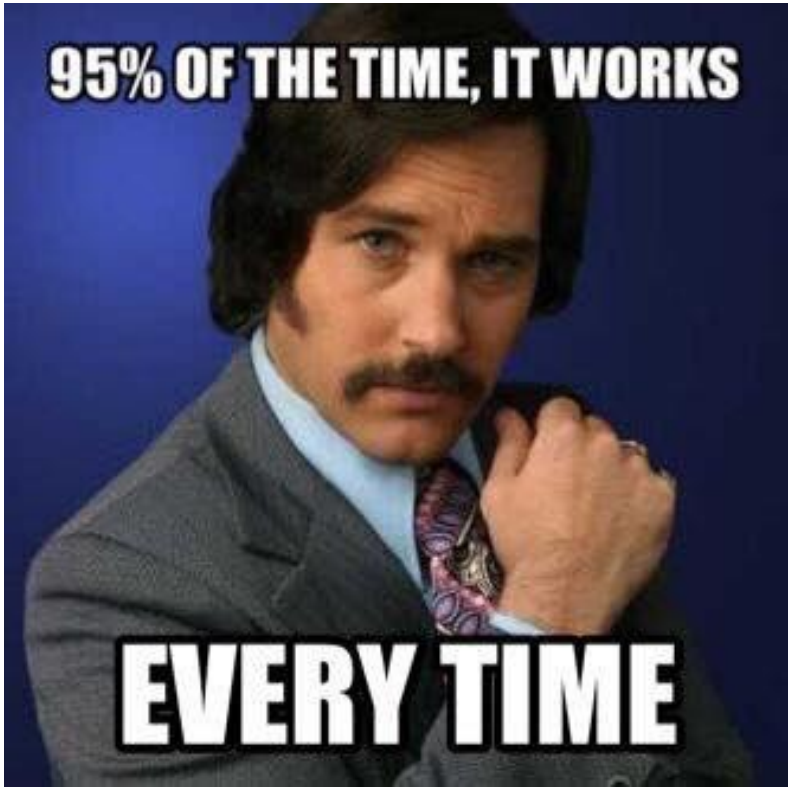
Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Derived Operators: Division

- Useful for queries like:

  "Find the sailors who have reserved **all** boats."



Derived Operators: Division

- Useful for queries like:

  "Find the sailors who have reserved **all** boats."

- Let *A* have 2 fields, *x* and *y* ; *B* only has 1 field *y* :
  - A/B = {<x> | ∀<y>∈ B, <x, y>∈A}
  - all x tuples (sailors) s.t. for **every** y tuple (boat) in B, there is an <x,y> tuple in A

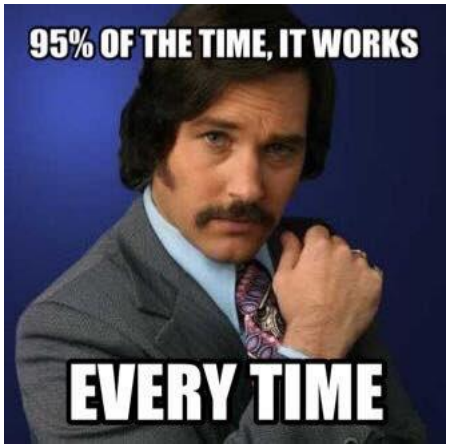
Generalized Definition:

Let:

- $R(A_1, A_2, ..., A_n, B_1, B_2, ..., B_m)$ be the dividend (the larger relation).
- $S(B_1, B_2, ..., B_m)$ be the divisor (the required set of values for B attributes).
- The result is $T(A_1, A_2, ..., A_n)$, which contains **all distinct values of** $A_1, A_2, ..., A_n$ **that are associated with every tuple in S**.

Formal Expression:

$$R(A_1, ..., A_n, B_1, ..., B_m) \div S(B_1, ..., B_m) = \{(a_1, ..., a_n) \in \pi_{A_1,...,A_n}(R) \mid \forall (b_1, ..., b_m) \in S, (a_1, ..., a_n, b_1, ..., b_m) \in R\}$$



Example

Examples of Division A/B

A/B = {<x> | ∀<y>∈ B, <x, y>∈A}

sid	bid
s1	b1
s1	b2
s1	b3
s1	b4
s2	b1
s2	b2
s3	b2
s4	b2
s4	b4

A

bid
b2

B1

bid
b2
b4

B2

bid
b1
b2
b4

B3

sid
s1
s2
s3
s4

A/B1

sid
s1
s4

A/B2

sid
s1

A/B3

Expressing A/B Using Basic Operators

- Can be equivalently expressed using basic operators
- **Idea:** For A/B, compute all x values that are not disqualified by some y value in B
  - x value is disqualified if by attaching y value from B, we obtain an <x,y> tuple that is not in A

Can you express this operator using basic operators?

A/B: $$\pi_x(A) - \underbrace{\pi_x((\pi_x(A) \times B) - A)}_{\text{Disqualified } x \text{ values}}$$

Example

Example of Division A/B

$$\pi_x(A) - \pi_x((\pi_x(A) \times B) - A)$$

sid	bid
s1	b1
s1	b2
s1	b3
s1	b4
s2	b1
s2	b2
s3	b2
s4	b2
s4	b4

A

bid
b2
b4

B

sid	bname
~~s1~~	~~b2~~
~~s1~~	~~b4~~
~~s2~~	~~b2~~
s2	b4
~~s3~~	~~b2~~
s3	b4
~~s4~~	~~b2~~
~~s4~~	~~b4~~

$\pi_x(A) \times B$ **-A**

sid
s1
s4

A/B

Quiz: Find names of sailors who have reserved a red OR green boat

- Hint:
  - Identify all red or green boats, then
  - find sailors who have reserved one of these boats

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
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22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Quiz: Find names of sailors who have reserved a red AND green boat

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
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22	101	10/10/98
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31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Quiz: Find names of sailors who have reserved a red AND green boat

?

$$\rho\ (Tempboats, (\sigma_{color='red' \wedge color='green'} Boats))$$

$$\pi_{sname}(Tempboats \bowtie Reserves \bowtie Sailors)$$

Quiz: Find the SIDs of sailors over age 20 who have not reserved a red boat

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
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Reserves

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22	104	10/7/98
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31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
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74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Quiz: Find the names of sailors who have reserved **all** boats

- Hint: Uses division; schemas of the input relations must be carefully chosen:

Sailors

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
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95	Bob	3	63.5

Reserves

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22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Summary

- Relational algebra: powerful!
- Has 5 basic operators
  - Selection σ
  - Projection π
  - Cross-product x
  - Set-difference -
  - Union U
- Can handle almost any query

Suggested Review

Exercises 4.1, 4.3, 4.5

Only RA required for the last two.

Next

Friday: Project 1, Joins, HW 1 solutions

Tuesday: Relational Calculus, SQL

$$\{ T \mid p(T) \}$$